

ServiceNow – SAP S4HANA RFC Execute RFC Action

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Scenarios

- In this scenario we will execute action on SAP S4 HANA RFC using "Execute RFC" spoke action.

Assumptions

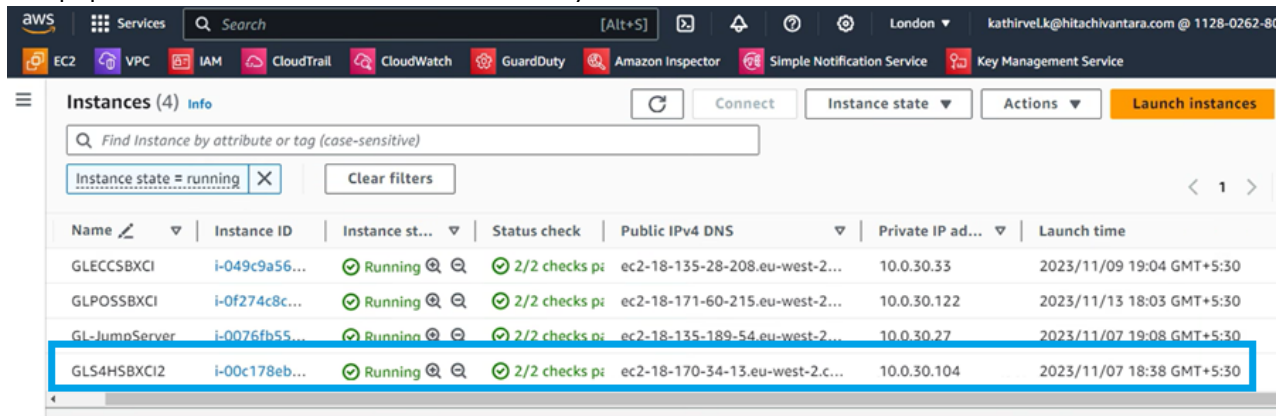
- ServiceNow S4HANA RFC Spoke is installed in the ServiceNow instance.
- Infrastructure: ServiceNow Integration Hub, and SAP S4 HANA RFC are all set-up.

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SAP Configuration (S4HANA RFC)

Map public URL with SAP S/4 HANA System that will be called from Service now



Name	Instance ID	Instance st...	Status check	Public IPv4 DNS	Private IP ad...	Launch time
GLECCSBXCI	i-049c9a56...	Running	2/2 checks pi	ec2-18-135-28-208.eu-west-2...	10.0.30.33	2023/11/09 19:04 GMT+5:30
GLPOSSBXCI	i-0f274c8c...	Running	2/2 checks pi	ec2-18-171-60-215.eu-west-2...	10.0.30.122	2023/11/13 18:03 GMT+5:30
GL-JumpServer	i-0076fb55...	Running	2/2 checks pi	ec2-18-135-189-54.eu-west-2...	10.0.30.27	2023/11/07 19:08 GMT+5:30
GLS4HSBXCI2	i-00c178eb...	Running	2/2 checks pi	ec2-18-170-34-13.eu-west-2.c...	10.0.30.104	2023/11/07 18:38 GMT+5:30

ServiceNow Integration Hub's SAP S4HANA RFC Spoke Configuration

Purpose: All actions are requesting change in the SAP system through the java connector.

Step 1 – ServiceNow: Create Credential Object

1. Provide Username and password for S4HANA RFC endpoint connection. (this will be provided by SAP team)

The screenshot shows the 'Basic Auth Credentials' form in ServiceNow. The form is titled 'SAP S4 HANA RFC Cred'. It has fields for 'Name' (SAP S4 HANA RFC Cred), 'Order' (100), 'User name' (RFC_USER), 'Password' (masked with dots), and 'Active' (checked). Below the form is a blue informational bar: 'A connection alias resolves your connection and credential at runtime. More than one Credential can be active per Connection Alias at a time. If more than one credential is active, they will be used in order.' Below the bar is a 'Credential alias' field with a lock icon. At the bottom, there are 'Update' and 'Delete' buttons. Below the form is a table titled 'Discovery IP Affinity' with a search bar and a 'New' button. The table has columns: MID server, Credential ID, IP address, IP address decimal, Domain, and Type. The table is empty, showing 'No records to display'.

MID server	Credential ID	IP address	IP address decimal	Domain	Type
No records to display					

Step 2 – ServiceNow: Create Connection in SAP_S4HANA_RFC connection alias

1. Go to Connection & Credential Aliases
2. Open alias SAP_S4HANA_RFC
3. Create new connection as following: choose credentials created in previous step, provide host that serves as endpoint for S4HANA connection, check use mid server option.

Connection
SAP S4 HANA RFC Connection 2

* Name: SAP S4 HANA RFC Connection 2

Credential: SAP S4 HANA RFC Cred

Active: ☒

Domain: global

A connection alias resolves your connection and credential at runtime. Only one Connection is active per Connection Alias at a time. [More Info](#)

* Connection alias: sn_sap_s4_hana_rfc.SAP_S4HANA_

* Host: ec2-18-170-34-13.eu-west-2.compute.a

Use MID server: ☒

Override default port: 3201

Advanced MID Server Configuration

MID Selection: Specific MID Server

* MID Server: sapmidlnx

Update Delete

Step 3 – ServiceNow: Enable Connection (optional)

If there are multiple connections under the same alias, make sure that newly created connection is active.

Connection & Credential Aliases
sn_sap_s4_hana_rfc.SAP_S4HANA_RFC

This record is in the [SAP S4 HANA RFC Spoke application](#), but [Global](#) is the current application. To edit this record click [here](#).

Name: SAP_S4HANA_RFC

ID: sn_sap_s4_hana_rfc.SAP_S4HANA_RFC

Application: SAP S4 HANA RFC Spoke

Type: Connection and Credential

Connection type: Basic

Support Multiple Active Connections: ☐

Default Retry Policy:

Configuration Template:

Select a template for configuring connection and credential records.

Related Links

[Create New Connection & Credential](#)

Connections (3) | Connection Attributes (8) | Child Aliases

Name Search

Actions on selected rows... New

Connection alias = sn_sap_s4_hana_rfc.SAP_S4HANA_RFC

Name	Connection type	Active	Credential	Use MID server	Protocol	Host	Override default port	Updated	Updated by
SAP S4 HANA RFC BTP Connection	Connection	false	S4 Hana BTP	true		https://integration-suite-vduzfeoz.it-cp...		2024-02-02 05:33:21	admin
SAP S4 HANA RFC Connection	Connection	false	SAP S4 HANA RFC Cred	true		18.170.34.13		2023-11-21 11:33:40	admin
SAP S4 HANA RFC Connection 2	Connection	true	SAP S4 HANA RFC Cred	true		ec2-18-170-34-13.eu-west-2.compute.amazo...	3201	2024-02-02 05:33:28	admin

1 to 3 of 3

Test Case 1: Execute RFC Action (Materials management)

ServiceNow will trigger an “execute RFC” request message using Integration Hub's SAP ECC RFC Spoke. The request message will be passed to the ECC RFC interface using the java connector. After the response is retrieved, it will be handled back to ServiceNow.

In this case custom RFC action is configured to be executed via action.

Step 1 - ServiceNow: Configure Action Execute RFC in Workflow

1. Create workflow with name e.g. S4 HANA RFC - Execute RFC action
2. Add action from S4 HANA RFC Spoke: “Execute RFC”
3. Configure action as following:
 - a. Connection Alias: SAP_S4HANA_RFC
 - b. RFC: Materials management – BAPI_MATERIAL_GET_DETAIL
 - c. Importing and changing Tab Inputs:
 - i. Material [MATERIAL]: AA123
 - ii. Plant: 9000
 - iii. Enqueue: false

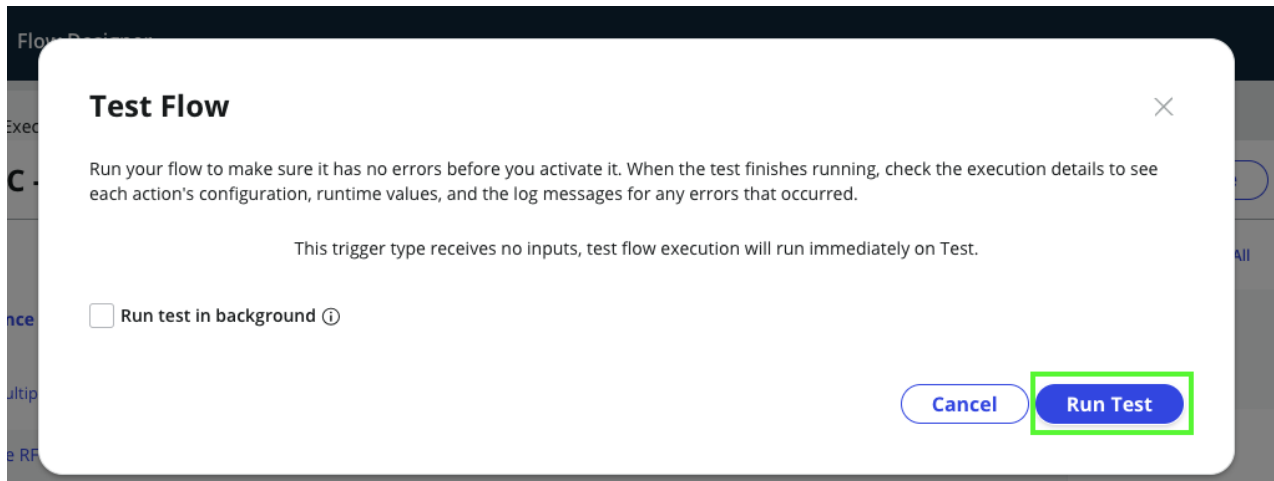
The screenshot shows the configuration interface for the 'Execute RFC' action in ServiceNow Flow Designer. The action is selected from a dropdown menu. Below it, the configuration fields are as follows:

- * Connection Alias [Connection & Credential Aliases]:** A dropdown menu showing 'sn_sap_s4_hana_rfc.SAP_S4HANA_RFC'.
- * RFC [SAP S/4HANA - RFC]:** A dropdown menu showing 'Material management - BAPI_MATERI...'.
- Import and Changing Tab Inputs:**
 - Material [MATERIAL]:** A text input field containing 'AA123'.
 - Plant:** A text input field containing '9000'.
- Tables Tab Inputs:** A section with a '+ Add field value' button.
- * Enqueue/Dequeue:** A dropdown menu showing 'false'.

At the bottom right, there are three buttons: 'Delete', 'Cancel', and 'Done'.

Step 2 – ServiceNow: Execute Workflow and Inspect

1. Open Flow Designer
2. Open workflow created in previous step
3. Click on Test Workflow
4. Click on Run Test



Step 3 – ServiceNow: Inspect Request Payload

1. After execution is completed, open execution details
2. Open Step 2 execution details and inspect payload



Step 2 - Ping and Execute RFC [Script]

Completed

Integration Metadata

Initial MID Server Selection

sapmidlnx

Step Configuration

VARIABLE NAME	RUNTIME VALUE	CONFIGURATION
Connection Alias	SAP_S4HANA_RFC @	sys_id
Required Runtime	mid	mid
Script	(function execute(inputs, outputs) { try { var credProvFactory = Packages.com.snc.common...	(function execute(inputs, outputs) { try { var credProvFactory = Packages.com.snc.common...
Select MID Server Using	use_connection_alias	use_connection_alias
conn_host	ec2-18-170-34-13.eu-west-2.compute.amazonaws.com	conn_host
conn_params	{"u_host":"","load_balance":"","u_group":"","u_message_server":"","r3nam e":"","u_client":"000","u_sy...	conn_params
cred_id	0fbcd1072fe2f55047578d1df699b6e4	cred_id
flag	false	enq_deq
op_type		operation_type
payload	("MATERIAL":"AA123","PLANT":9000)	json_in
rfc	BAPI_MATERIAL_GET_DETAIL	rfc_name

• Step 4 – SAP Execution

The ServiceNow endpoint triggers a call to the SAP Remote Function Call (RFC) Module, and the response comprises the return parameters of the same module.

Step 5 – ServiceNow: Inspect Response Payload

1. Open execution details
2. Inspect output data section – it will contain material plant details in returned object

 Execute RFC		Completed
Configuration Details		
VARIABLE NAME	RUNTIME VALUE	CONFIGURATION
Connection Alias	SAP_S4HANA_RFC	SAP_S4HANA_RFC
RFC	BAPI_MATERIAL_GET_DETAIL	BAPI_MATERIAL_GET_DETAIL
Import and Changing Tab Inputs	MATERIAL=AA123^PLANT=9000	MATERIAL=AA123^PLANT=9000
Tables Tab Inputs		
Response	{ "children": [{ "children": [{ "name": "PUR_GROUP", "label": "Purchasing Group", "type": "string" }, { "name": "I...	{ "children": [{ "children": [{ "name": "PUR_GROUP", "label": "Purchasing Group", "type": "string" }, { "name": "I...
Enqueue/Dequeue	false	false
Output Data		
VARIABLE NAME	RUNTIME VALUE	CONFIGURATION
Action Status	{ "Action Status": { "code": 0, "message": "Success" } }	
Don't Treat as Error	true	true
Error Message		err
JSON Reponse	{ "RETURN": { "MESSAGE_V2": "", "MESSAGE_V3": "", "CODE": "", "MESSAGE_V4": "", "LOG_NO": "", "MESSA...	response
Response	<pre> { "MATERIALPLANTDATA": { "ISSUE_UNIT": null, "PUR_GROUP": "009" }, "MATERIALVALUATI... </pre>	response
Status	success	status

1. You can explore further by clicking on the value.

Viewing output [dynamic_object]



Rendered HTML

Raw Text

Code

```
1 {  
2   "MATERIALPLANTDATA" : {  
3     "ISSUE_UNIT" : null,  
4     "PUR_GROUP" : "009"  
5   },  
6   "MATERIALVALUATIONDATA" : {  
7     "CURRENCY" : null,  
8     "CURRENCY_ISO" : null,  
9     "MOVING_PR" : "0",  
10    "PRICE_CTRL" : null,  
11    "PRICE_UNIT" : "0",  
12    "STD_PRICE" : "0"  
13  },  
14  "MATERIAL_GENERAL_DATA" : {  
15    "BASE_UOM" : "EA",  
16    "BASE_UOM_ISO" : "EA",  
17    "BASIC_MATL" : null,  
18    "BASIC_MATL_NEW" : null,  
19    "CHANGED_BY" : "SRAMALINGAM",  
20    "CONTAINER" : null,  
21    "CREATED_BY" : "SRAMALINGAM",  
22    "CREATED_ON" : "2023-12-07",  
23    "DIVISION" : null,  
24    "EAN_CAT" : null,  
25    "EAN_UPC" : null,  
26    "EMPTYBOM" : null,  
27    "GROSS_WT" : "0",
```

Close

Step 6 – Validation on SAP Side

Display Material Document 5000000012 - SARAVANAN SRAMALINGAM

Display Material Document 5000000012 - SARAVANAN SRAMALINGAM

Show Overview | Hold | Check | Post | Help

ACH Display | R02 Material Document | 5000000012 | 2023

General | Vendor | Doc. info

Document Date: 11.12.2023 | Delivery Note: | Vendor: Hitachi Vantara
 Posting Date: 11.12.2023 | Bill of Lading: | HeaderText: |
☐ 1 Individual Slip

Line	Mat. Short Text	Qty. in Unit	UoM	Stoc	Batch	Valuation type	M...	D	Stock Type	Plnt	S...	Customer
1	IC Engine	100	EA	STORAGE LC1			101		Unrestrict...	GL plant		

Material | Quantity | Where | Purchase Order Data | Partner

Purchase order: 4500000033 | 10 |
 Reference Document: 5000000012 | 1 |
 Item category: Standard
☒ DCI for Mat. Doc. Item
☒ Del. Compl. Ind. PO Item

Line: 1